```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("/content/netflix_titles.csv")

df.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	ra
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	P
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV

Next steps: Generate code with df | New interactive sheet

```
print("Shape of Dataset:", df.shape)

print("\nDataset Information:")
df.info()
```

```
Shape of Dataset: (8807, 12)

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
```

```
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

```
print(df.isnull().sum())
```

```
show_id            0
type               0
title              0
director        2634
cast             825
country          831
date_added        10
release_year       0
rating             4
duration           3
listed_in          0
description        0
dtype: int64
```

```
df['director'] = df['director'].fillna('Unknown')
df['cast'] = df['cast'].fillna('Unknown')
df['country'] = df['country'].fillna('Unknown')
df['rating'] = df['rating'].fillna('Unknown')
df['duration'] = df['duration'].fillna('Unknown')


df.dropna(subset=['date_added', 'release_year'], inplace=True)

print("Missing values after cleaning:")
print(df.isnull().sum())
```

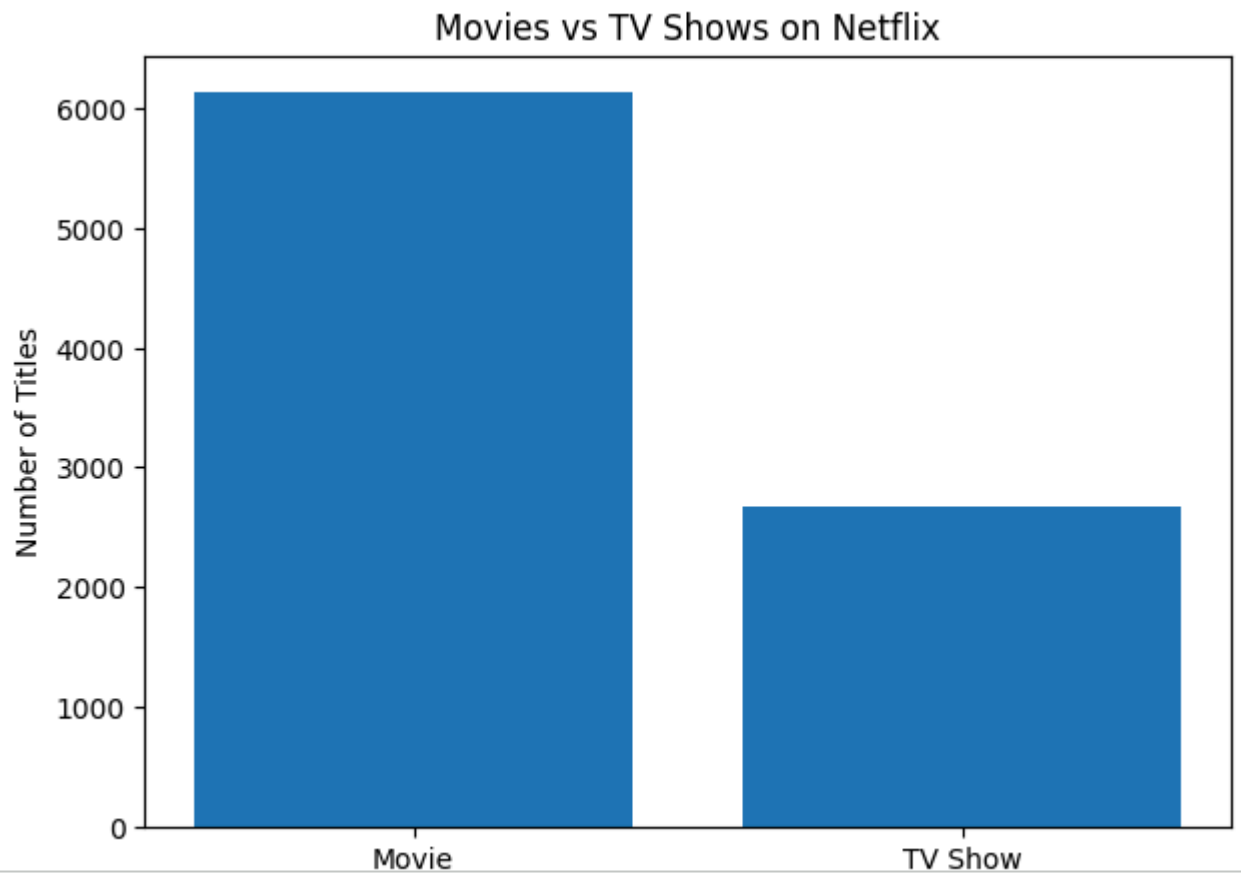
```
Missing values after cleaning:
show_id         0
type            0
title           0
director        0
cast            0
country         0
date_added      0
release_year    0
rating          0
duration        0
listed_in       0
description     0
dtype: int64
```

```
type_count = df['type'].value_counts()

plt.figure(figsize=(7,5))
plt.bar(type_count.index, type_count.values)

plt.title("Movies vs TV Shows on Netflix")
plt.xlabel("Type")
plt.ylabel("Number of Titles")

plt.show()
```




```
df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')

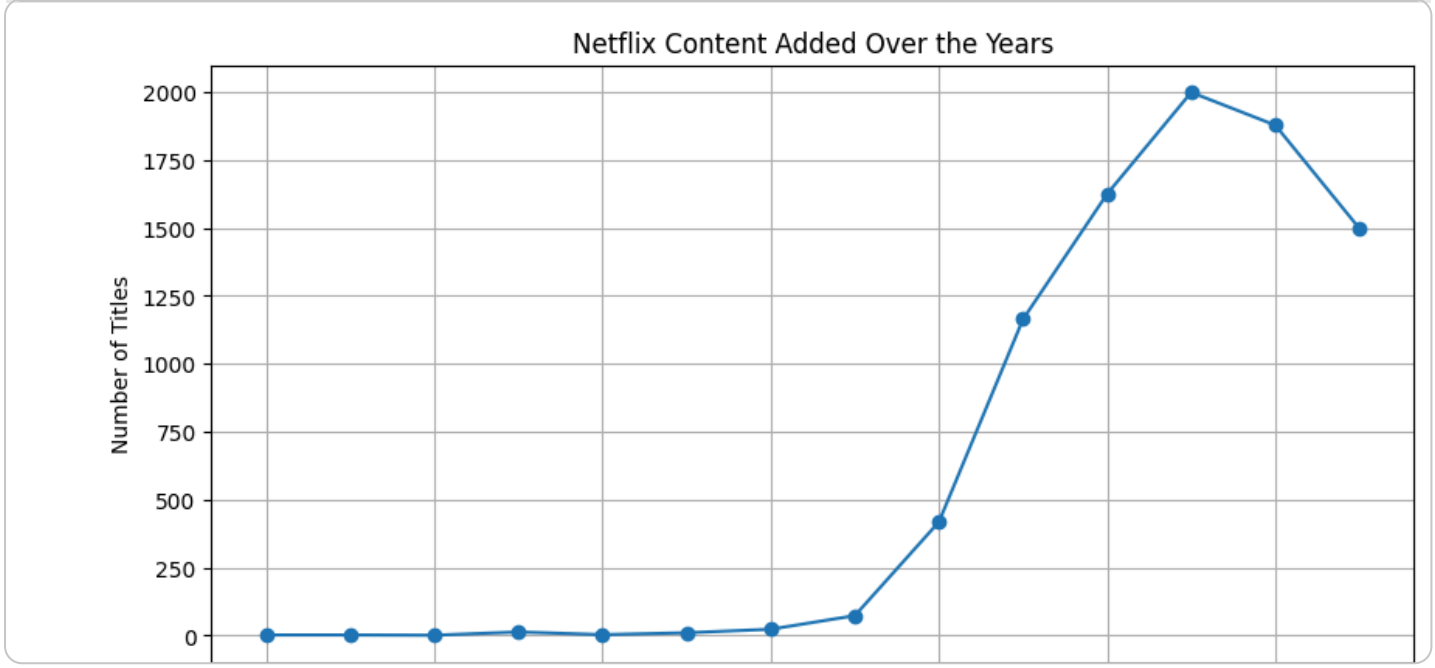
df['year_added'] = df['date_added'].dt.year

year_count = df['year_added'].value_counts().sort_index()

plt.figure(figsize=(10,5))
plt.plot(year_count.index, year_count.values, marker='o')

plt.title("Netflix Content Added Over the Years")
plt.xlabel("Year")
plt.ylabel("Number of Titles")

plt.grid()
plt.show()
```

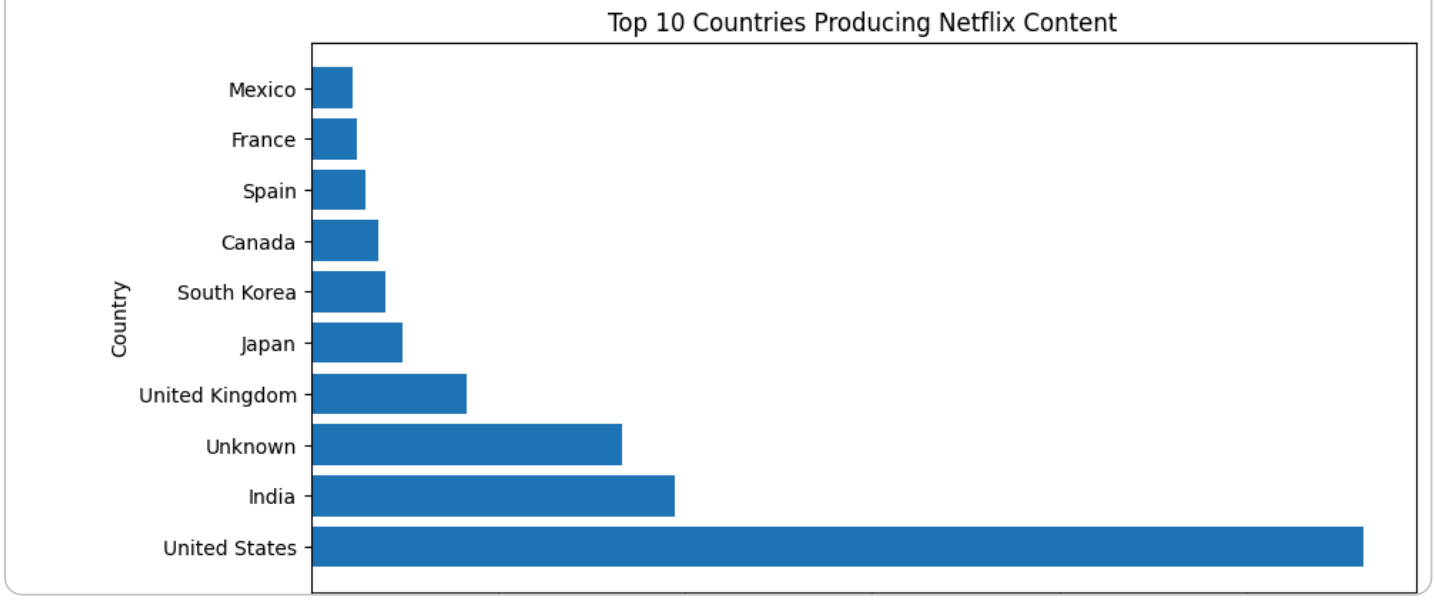



```
country_count = df['country'].value_counts().head(10)

plt.figure(figsize=(10,5))
plt.barh(country_count.index, country_count.values)

plt.title("Top 10 Countries Producing Netflix Content")
plt.xlabel("Number of Titles")
plt.ylabel("Country")

plt.show()
```




```
genres = df['listed_in'].str.split(', ').explode()

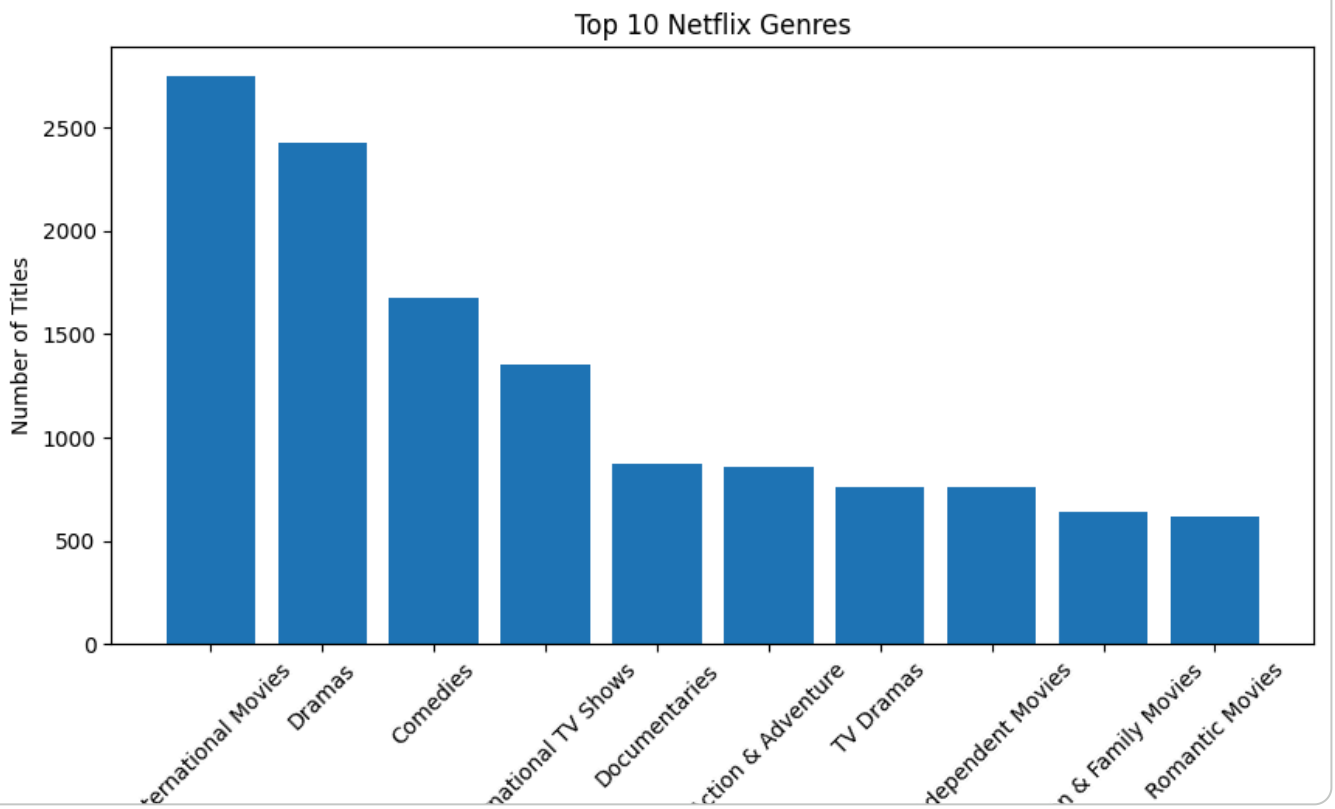
top_genres = genres.value_counts().head(10)

plt.figure(figsize=(10,5))
plt.bar(top_genres.index, top_genres.values)

plt.title("Top 10 Netflix Genres")
plt.xlabel("Genre")
plt.ylabel("Number of Titles")

plt.xticks(rotation=45)
```

```
plt.show()
```



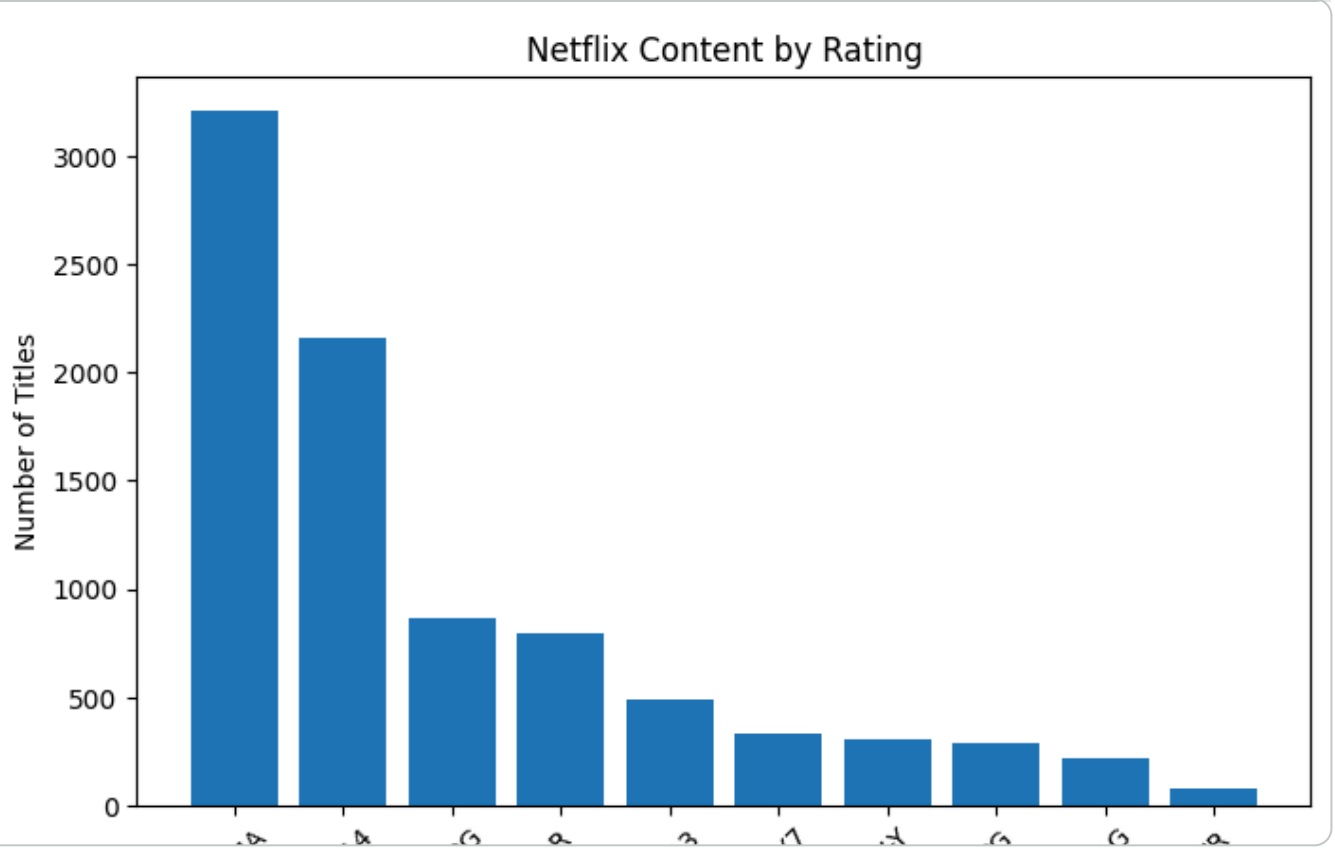

```
rating_count = df['rating'].value_counts().head(10)

plt.figure(figsize=(8,5))
plt.bar(rating_count.index, rating_count.values)

plt.title("Netflix Content by Rating")
plt.xlabel("Rating")
plt.ylabel("Number of Titles")

plt.xticks(rotation=45)

plt.show()
```

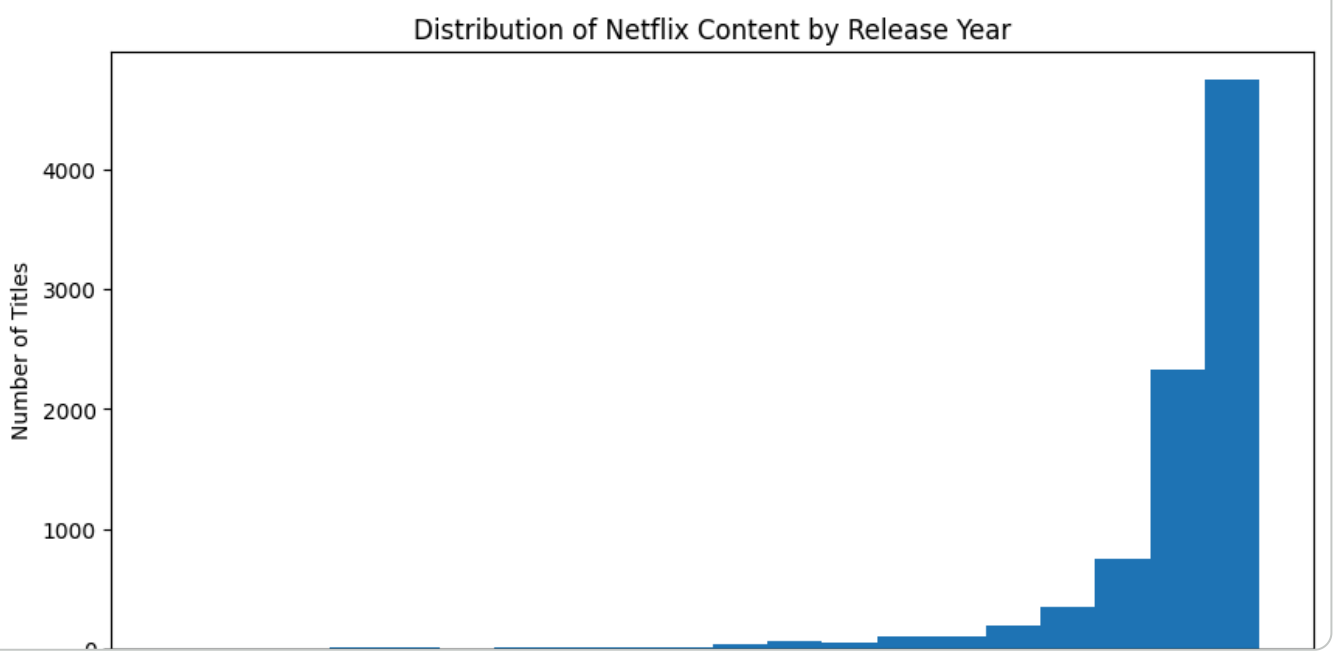



```
plt.figure(figsize=(10,5))

plt.hist(df['release_year'], bins=20)

plt.title("Distribution of Netflix Content by Release Year")
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")

plt.show()
```




```
numeric_data = df.select_dtypes(include=np.number)

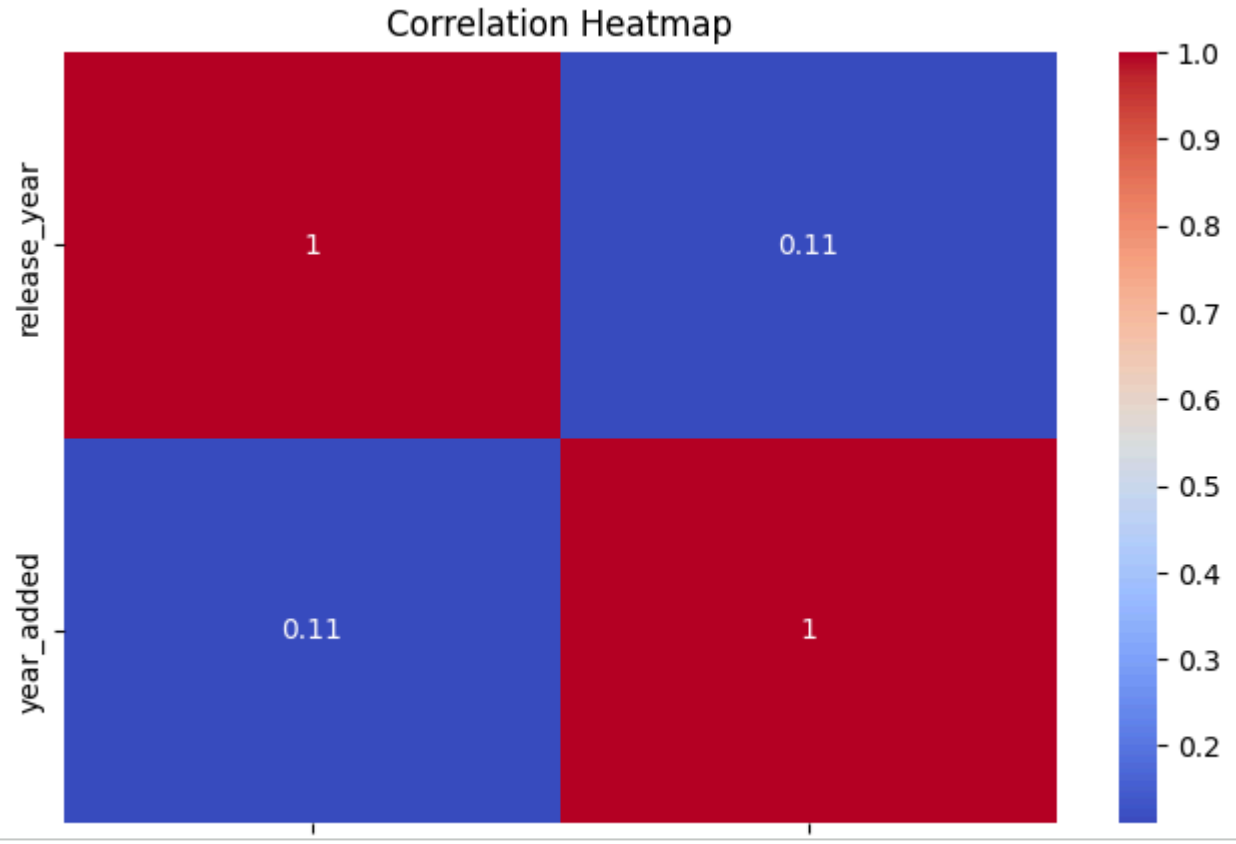
plt.figure(figsize=(8,5))

sns.heatmap(
    numeric_data.corr(),
    annot=True,
```

```
    cmap="coolwarm"
)

plt.title("Correlation Heatmap")

plt.show()
```




```
total_titles = len(df)

total_movies = len(df[df['type'] == 'Movie'])

total_tv_shows = len(df[df['type'] == 'TV Show'])

unique_countries = df['country'].nunique()

unique_genres = genres.nunique()

print("Total Titles:", total_titles)
print("Total Movies:", total_movies)
print("Total TV Shows:", total_tv_shows)
print("Countries:", unique_countries)
print("Genres:", unique_genres)
```

```
Total Titles: 8797
Total Movies: 6131
Total TV Shows: 2666
Countries: 749
Genres: 42
```

```
kpis = {
    "Total Titles": total_titles,
    "Movies": total_movies,
    "TV Shows": total_tv_shows,
    "Countries": unique_countries,
    "Genres": unique_genres
```

```
}

for key, value in kpis.items():
    print(f"{key}: {value}")
```

```
Total Titles: 8797
Movies: 6131
TV Shows: 2666
Countries: 749
Genres: 42
```

```
print("DATA STORY")

print("\n1. Netflix contains both Movies and TV Shows.")
print("2. Movies generally represent a larger portion of the Netflix catalog.")
print("3. Netflix content has increased significantly over the years.")
print("4. A few countries contribute a large amount of Netflix content.")
print("5. Drama and International content are among the major genres.")
print("6. Netflix contains content across multiple audience ratings.")
```

```
DATA STORY

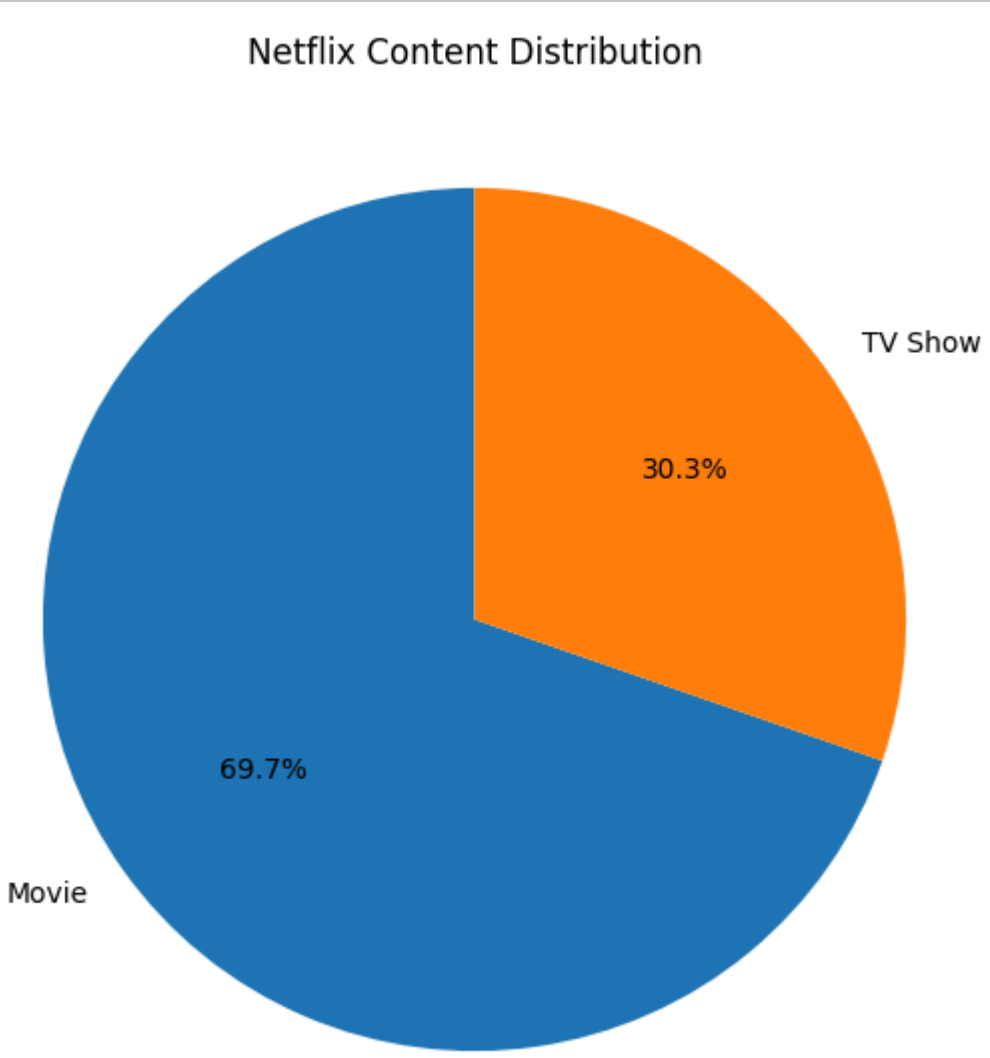
1. Netflix contains both Movies and TV Shows.
2. Movies generally represent a larger portion of the Netflix catalog.
3. Netflix content has increased significantly over the years.
4. A few countries contribute a large amount of Netflix content.
5. Drama and International content are among the major genres.
6. Netflix contains content across multiple audience ratings.
```

```
plt.figure(figsize=(7,7))

plt.pie(
    type_count.values,
    labels=type_count.index,
    autopct='%1.1f%%',
    startangle=90
)

plt.title("Netflix Content Distribution")

plt.show()
```




```
print("""
DATA STORY:

Netflix has a large and diverse content library consisting of
Movies and TV Shows.

The analysis shows that Movies form a significant part of the
Netflix catalog. The amount of content added to Netflix has
also changed considerably over the years.

A small number of countries contribute a large share of the
available content. Different genres and ratings help Netflix
target different groups of viewers.

Business Recommendation:
Netflix can use these insights to identify popular genres,
target specific countries, understand audience preferences,
and plan future content investments.
""")
```

```
DATA STORY:

Netflix has a large and diverse content library consisting of
Movies and TV Shows.

The analysis shows that Movies form a significant part of the
```

```
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```

Part - B

1. What is data storytelling? How is it different from data visualization? Data storytelling is the process of using data, visuals, and a narrative to explain insights. Data .
2. Why is storytelling important in business analytics and decision-making? It makes complex data easy to
3. What are the essential components of an effective data story? The main components are:
4. How do Key Performance Indicators (KPIs) enhance business reporting? KPIs provide measurable values to track business performance.